

Responsible Anthropomorphic Design in LLM-Based Chatbots: Bridging Practitioner Practice and Children's Wellbeing

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Abstract

My doctoral research explores how practitioners can design anthropomorphism in large language model (LLM)-based chatbots to protect children's wellbeing. Anthropomorphism is the cognitive process through which people attribute human characteristics, intentions, and mental states to non-human entities, a process that LLM-based chatbots actively trigger child users to do so through human-like language, emotional responsiveness, and expressed preferences. Through a systematic literature review, my completed dissertation work establishes that these anthropomorphic design features simultaneously support and undermine children's wellbeing through the identical psychological pathway, with consequences varying across developmental stages. My proposed future work will first explore how practitioners currently conceptualize children's wellbeing and what individual and structural conditions shape their anthropomorphic design practice. I will then co-design with practitioners to produce a taxonomy of design tensions and actionable design principles grounded in developmental evidence and practitioner reality. My dissertation aims to contribute to the Child-Computer Interaction (CCI) field by explaining how anthropomorphic design affects children's wellbeing, how practitioners currently navigate these decisions in practice, and what design principles can guide responsible anthropomorphic design in LLM-based chatbots.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing**; **Human computer interaction (HCI)**.

Keywords

Child wellbeing, LLM-based chatbot, Anthropomorphism

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1 Introduction

Among artificial intelligence (AI) technologies, large language models (LLM)-based chatbots have entered children's educational, therapeutic, and recreational lives at an unprecedented scale [29]. Children are among the fastest-growing groups of AI users, with many incorporating LLM-based chatbots into their daily routines [4, 34]. Across a growing range of domains, researchers and designers have deployed these systems to support children in education [25], creative coding [12], collaborative storytelling and reading [10, 15], environmental awareness [36], mental health [26], family therapy [24], and pediatric healthcare communication [32].

LLMs have transformed chatbots by embedding social responsiveness that triggers anthropomorphism, a psychological phenomenon where human qualities are attributed to non-human entities [5, 13, 21]. Children are naturally susceptible to this [20], with developmental stages and deliberate human-like design further activating this predisposition [3, 35, 38]. Severe interactions have been linked to self-harm [2, 9, 39]. The research community has addressed this through child safety, defined as the prevention of acute and identifiable harm [1, 30], by implementing risk taxonomies [18, 29] and benchmarks for harmful outputs [19, 23, 31, 33]. However, child safety measures remain distinct from the broader support of child wellbeing.

Child wellbeing refers to the quality of a child's life across multiple simultaneous dimensions [6], meaning not simply the absence of harm but the active presence of conditions for flourishing [11, 17]. Research on children's digital wellbeing remains fragmented and dominated by deficit framings focused on problematic use, screen time, and harm avoidance rather than on the positive developmental conditions flourishing requires [8, 16]. Huppert [17]'s wellbeing model, which encompasses positive relationships, resilience, competence, meaning, and self-determination, offers a richer normative standard, but no existing framework operationalizes it at the developmental trajectory level. This means that accumulative effects, such as gradual relationship displacement or the formation of inaccurate mental models of social agency, remain empirically invisible. Critically, anthropomorphism as the specific mechanism through which AI design activates these developmental consequences in children has received almost no systematic attention, leaving the children's wellbeing implications of anthropomorphic design unexamined.

This gap extends to the practitioners who build these systems. Responsible AI practitioner research has reproduced the same deficit framing, on harm anticipation and mitigation through safety-oriented frameworks [7] and foresight tools [37], while treating adult or general user populations as the implicit standard. To the

best of my knowledge, only one empirical study has examined how practitioners operationalize child wellbeing as a responsible design concept, found that practitioners default to engagement metrics without a conceptual framework for connecting design decisions to developmental consequences [22]. Whether this pattern is compounded when the technology is specifically designed to behave anthropomorphically remains unknown. Value Sensitive Design [14], which embeds values constitutively into the design process, provides the theoretical bridge this dissertation uses to connect developmental evidence to responsible practice, yet no co-designed framework currently exists that applies it to anthropomorphic design for children specifically. Together these gaps in developmental evidence, in wellbeing frameworks, and in practitioner guidance motivate this dissertation. My dissertation, therefore, asks: **how can practitioners design anthropomorphism in LLM-based chatbots to protect children’s wellbeing?** Three sequential studies address this question:

- **Study 1:** A systematic review of anthropomorphic drivers and their consequences for children’s wellbeing across developmental stages
- **Study 2:** A semi-structured interview study exploring how practitioners conceptualize and navigate anthropomorphic design decisions in practice in relation to children’s wellbeing
- **Study 3:** A co-design workshop study producing design tensions and principles for responsible anthropomorphic design for children’s wellbeing

2 Work To date

My dissertation follows a multi-method qualitative design across three studies. Study 1, a systematic literature review, is complete and accepted at ACM IDC 2026 conference.

2.1 Study 1: Understanding Anthropomorphism as a Double-Edged Sword in Child-LLM-Based Chatbot Interactions through a Systematic Literature Review (ACM IDC 2026)

Following PRISMA guidelines [27], I conducted a systematic review of 35 empirical studies published between 2022 and August 2025, drawn from ACM Digital Library, Scopus, and Web of Science. Analysis combined deductive application of the SEEK framework [13], which discusses the psychological determinants of anthropomorphism to organize anthropomorphic drivers with inductive thematic coding of outcomes, inspired by [17], wellbeing dimensions, and analyzed across Piagetian developmental stages [28].

Addressing what drives anthropomorphism in children’s LLM-based chatbot interactions, I identified four drivers embedded in the design of LLM-based chatbots, such as Human-Like Persona Construction, Supportive Companionship, Adaptive Scaffolding, and Non-Human Embodiment Design. The first three align with SEEK’s three determinants; the fourth extends the framework by showing that explicitly non-human fantasy characters paradoxically enhance rather than reduce anthropomorphic attribution. Addressing what the consequences are across developmental stages

and wellbeing dimensions, I identified five outcomes such as Social Ties Construction, Social Boundary Exploration, Human Narrative Attribution, Paradoxical Social-Moral Responses, and Dual Consciousness Formation as shown in figure 1.

The findings suggest that pre-operational children (ages 2 to 6) face the highest vulnerability because LLM’s language fluency appears to reinforce animistic thinking beyond typical developmental timelines, yet this group remains the most underrepresented in the literature. Concrete operational children (ages 7 to 12) appear to represent an optimal but underutilized intervention window, combining peak anthropomorphism with emerging logical capacity that may support AI literacy education. Adolescents (ages 12 and above) exhibit a formal operational paradox where sophisticated technical knowledge of AI architecture coexists with romantic relationships, clinical dependency, and threats to self-worth.

The central finding is that anthropomorphic design functions as a double-edged sword indicating that the same anthropomorphic design feature in LLM based chatbots produces benefit and developmental risk through the identical psychological pathway. The reviewed evidence suggests that features enabling emotional support and learning scaffolding simultaneously contribute to patterns consistent with resilience erosion and inaccurate mental model formation, making harm-reduction approaches that treat features as either safe or unsafe insufficient. Landesman et al. [22] confirm that practitioners lack conceptual vocabulary for this problem, defaulting to engagement metrics and safety checklists. This finding directly motivates Study 2.

3 Current and Future Dissertation Work

Studies 2 and 3 build sequentially on Study 1, exploring practitioner practice and co-producing design principles respectively.

3.1 Study 2: Anthropomorphic Design Practice and Children’s Wellbeing: A Practitioner-Side Exploration

Addressing Study 2’s research question introduced above, I will conduct semi-structured interviews with 18 to 20 practitioners across four role types, such as designers, product managers, AI engineers, and UX researchers, spanning startups and large organizations. Participants are not required to have worked on child-facing products, because individual and structural conditions shaping practice are properties of LLM chatbot development broadly. Each interview follows a two-phase protocol, where Phase 1 maps practitioners’ conceptualizations on their own terms without introducing Study 1’s findings, while Phase 2 probes individual orientations and structural conditions, including organizational priorities, performance metrics, and stakeholder pressures. Analysis will use inductive thematic analysis in Atlas.ti, followed by an analysis examining where practitioner design logic aligns with or departs from Study 1’s driver-outcome evidence. This output directly constitutes the input material for Study 3.

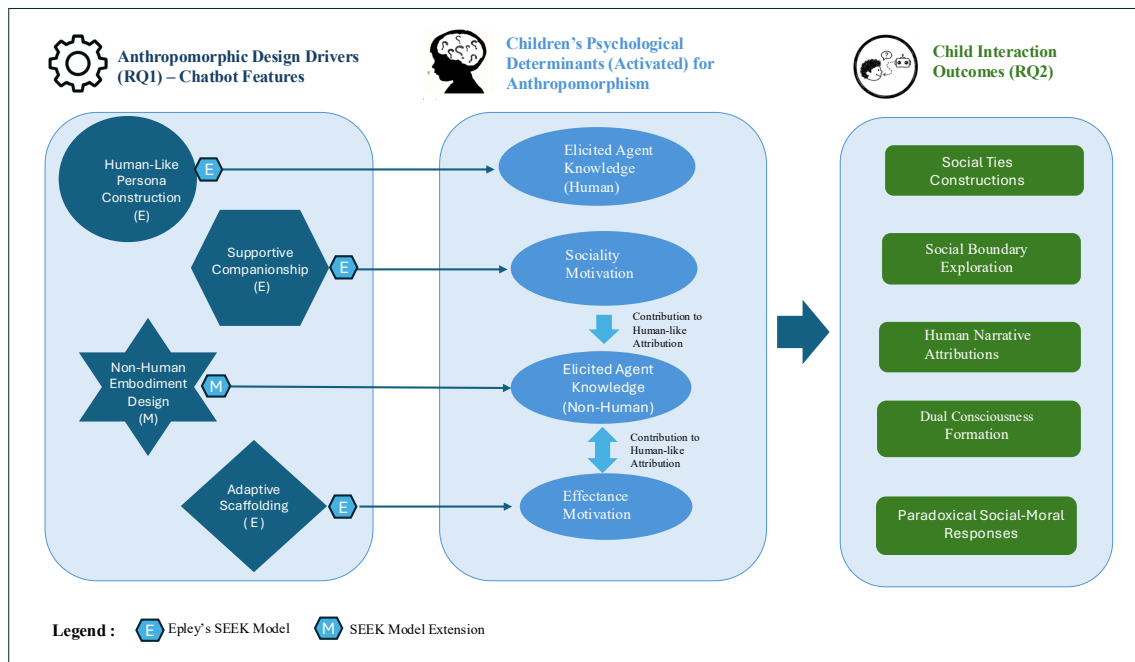


Figure 1: Activation of Psychological Determinants and Outcomes Associations.

3.2 Study 3: Co-Designing Responsible Anthropomorphic Design Principles

Addressing Study 3's research question introduced above, I will conduct co-design workshops grounded in Value Sensitive Design [14], which commits to embedding values constitutively into the design process rather than applying them as post-hoc constraints. I will recruit 18 to 20 practitioners across four to six sessions, with all four role types represented in each. Phase 1 is exploratory: practitioners map their design decisions onto Study 1-based design tension cards and Huppert [17]'s wellbeing framework to co-construct the tensions defining the responsible anthropomorphic design space. Phase 2 is generative: practitioners use the Phase 1 tension taxonomy to propose, debate, and refine design principles that embed children's wellbeing as a constitutive design value, consistent with VSD's technical investigation approach. Provocation materials include design tension cards from Study 1's driver-outcome taxonomy and anonymized practitioner vignettes from Study 2's analysis of where practitioner design logic aligns with or departs from developmental evidence, making that evidence and the organizational and structural barriers practitioners face actionable rather than merely informational. I expect Study 3 to generate a co-produced taxonomy of design tensions, a set of design principles grounded in all three studies simultaneously, and a boundary finding identifying which tensions require organisational or regulatory intervention beyond what design practice alone can achieve.

4 Current and Expected Outcomes

This dissertation contributes to the CCI community through (1) a systematic account of how anthropomorphic design features affect children's wellbeing across developmental stages, (2) an empirical exploration of how practitioners conceptualise and navigate these design decisions in practice, and (3) a co-produced set of design tensions and principles giving practitioners actionable guidance for responsible anthropomorphic design for children.

5 Positionality

I am a third-year, full-time Ph.D. candidate in the Information Technology program at the University of Cincinnati's School of Information Technology. The projected completion date of my doctoral degree will be around May 2027 or summer 2027. I have never attended a doctoral consortium at any previous ACM conference. The themes of my research are HCI, CCI, Responsible AI, and, specifically, responsible anthropomorphic design in LLM-based chatbots.

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